

Technical Data Sheet

Date : 01.07.2025
Product Name : Transformer Oil (Unused inhibited high grade insulating oil) - TRANSOL HGX
Grade : IEC - 60296 (Edition 5.0 2020-06) - Inhibited, Type A (fully Inhibited high grade oils) & Customer Spec.
Customer Name : PGCIL Rev. No.01

Sr. No.	Property	Test Method	Limits	Typical Values
Function :				
1a	Viscosity at 40 °C	ISO 3104	Max. 12 mm ² /s	10.8
1b	Viscosity at - 30 °C	ISO 3104	Max. 1800 mm ² /s	1100
2	Appearance	A representative sample of the oil shall be examined in a 100mm thick layer, at ambient temperature	The oil shall be clear and bright , transparent and free from suspended matter or sediment	Clear , free from sediment and suspended matter
3	Pour Point	ISO 3016	Max. - 40 °C	- 42
4	Water Content	IEC 60814	Max. 30 mg / kg - for bulk supply Max. 40 mg /kg - delivery in drums/IBC	12 14
5	Breakdown voltage	IEC 60156	Min. 50 kV as delivered Min. 70 kV after laboratory treatment	60 >72
6	Density at 20 °C	ISO 3675	0.820 - 0.895 g / ml	864
	Density at 29.5 °C	ISO 3675	-	859
7	DDF at 90 °C & 40 to 60 Hz	IEC 60247	Max. 0.0025	0.00035
8	Negative Impulse testing, kVp at 25°C	ASTM D-3300	Min. 145	178
9	Carbon Type composition (% of Aromatic, Paraffins and Naphthenic Compounds)	IEC 60590 & IS 13155 (FTIR)	Aromatic 4 - 12 % Paraffins < 50 % Balance Shall Be Naphthenic Compounds	4.50 48.50 47.00
2 - Refining / Stability				
1	Colour	ISO 2049	L 0.5 (less than 0.5)	L 0.5
2	Appearance	-	Clear, free from sediment and suspended matter	Clear, free from sediment & suspended matter
3	Acidity	IEC 62021-2 or 62021-1	Max. 0.01 mg KOH/g	0.0014
4	Interfacial Tension at 27 °C	IEC 62961 or ASTM D 971	Min. 0.043 N/m	0.047
5	Total sulphur content	ISO 8754 / or ISO 20847	Max. 0.05 % (before oxidation test)	0.002
6	Corrosive Sulphur	DIN 51353	Not corrosive	Non corrosive
7	Potentially Corrosive Sulphur	IEC 62535	Not corrosive	Non corrosive
8	DBDS	IEC 62697-1	Not detectable (< 5 mg/kg)	Non detectable
9	Inhibitors of IEC 60666	IEC 60666	Inhibited Oil (I); 0.08 % to 0.4 %	0.38
10	Metal passivator additives of IEC 60666	IEC 60666	Not detectable (< 5 mg/kg) or as agreed upon with the purchaser	Not detectable
11	Other additives	-	-	NIL (Not added)
12	2 - Furfural & related compounds content	IEC 61198	Not detectable (< 0.05 mg/kg) for each individual compound	Not detectable
13	Stray gassing under thermo - oxidative stress	Procedure in Clause A.4 of IEC 60296-2020 (oil in the saturated with air) presence of copper	Non stray gassing < 50 µl/l of Hydrogen (H ₂) < 50 µl/l methane (CH ₄) < 50 µl/l ethane (C ₂ H ₆)	Non stray gassing Non stray gassing Non stray gassing
3 - Performance :				
1	Oxidation Stability (*values at the end of oxidation test)	IEC 61125 Test duration : (I) Inhibited Oil : 500 h		
a)	Total acidity*	4.8.4 of IEC 61125: 2018	Max. 0.3 mg KOH / gm	0.14
b)	Sludge*	4.8.1 of IEC 61125: 2018	Max. 0.05 %	0.012
c)	DDF at 90 °C*	4.8.5 of IEC 61125: 2018	Max. 0.05	0.0042
4 - Health , safety and environment :				
1	Flash point	ISO 2719	Min. 135 °C	148
2	PCA content	IP 346	< 3.0 %	0.52
3	PCB content	IEC 61619 / ASTM D 4059	Not detectable (< 2 mg/kg)	Not detectable


Authorised Signatory